

Microprocessor & Computer Architecture

BCA — Semester II — 2022 — Answer Sheet

Subject Code: CACS155

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. What is the use of flag? Explain different flags of 8085 with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Draw the timing diagram of LXI H, 5070H instruction.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Explain how does control unit determines the type of instruction after it is read from memory.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is microprogram sequencer? Differentiate between hardwired control unit and microprogrammed control unit.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What are the basic differences between a branch instruction, a call subroutine instruction, and program interrupt?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What do you mean by pipelining hazards? Explain briefly.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short notes on: a) Addressing modes of 8085 b) Vector Processing

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions [2x10=20]

9. Define symbolic microprogram. You have a list of data stored at memory location starting at 8050H. The end of the data array is indicated by data byte 00H. Write a program to add the set of readings. Store the sum at 9050 and total carry at 9051H.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Write a program to evaluate the arithmetic statement: $X = A - B + C * (D * E + F) / C + H * K a$ Using a general register computer with two address instructions. b) Using an accumulator type computer with one address instructions. c) Using a stack organized computer with zero-address operation instructions.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define Control Word. Explain in detail about the structure of control unit and its basic requirement.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.